System and Devices (SYS 3)

Lecture 8: ILP – Compiler Techniques

Dr Pengcheng Liu

Introduction

- Pipelining overlaps execution of instructions
- Exploits Instruction Level Parallelism (ILP)
- Two main approaches:

  - Compiler-based static approaches

  - Hardware-based dynamic approaches
- **The goal of exploiting ILP: Minimizing CPI**

  - Pipeline CPI = Ideal (based) CPI + Structural stalls + Data hazards stalls + Control stalls

Parallelism Limitation with Basic Block

- The basic block – a straight-line code sequence without branches in except to the entry and no branches out except at the exit.
- Parallelism with basic block is limited. Typical size of basic block few instructions only. Must optimize across multiple blocks (branches)

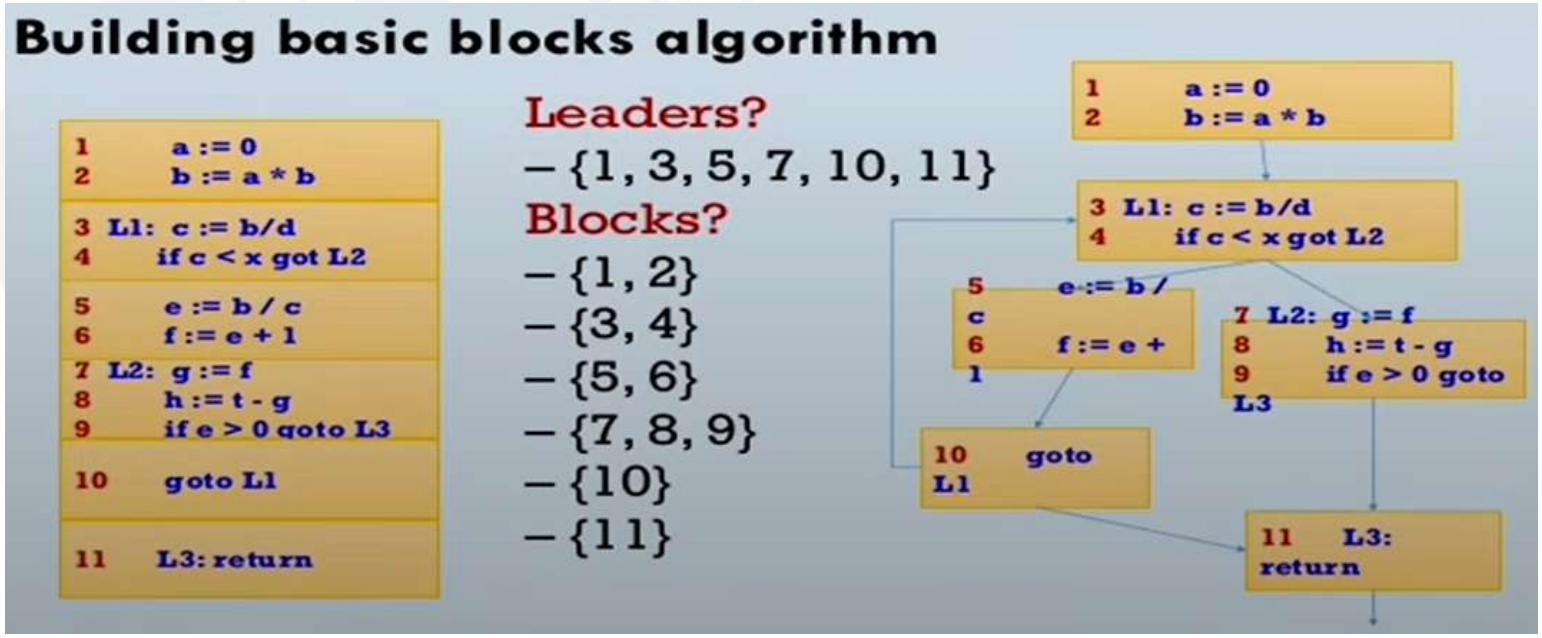
```
w = 0;
x = x + y;
y = 0;
if( x > z)
  {
    y = x;
    x++;
  }
else
  {
    y = z;
    z++;
  }
w = x + z;
```

```
w = 0;
x = x + y;
y = 0;
if( x > z)

y = x;
x++;

y = z;
z++;

w = x + z;
```



Data Dependence

- Loop-Level Parallelism

  - Unroll loop statically or dynamically
- Challenges: Data dependency
- Data dependence conveys possibility of a hazard
- Dependent instructions cannot be executed simultaneously
- Pipeline determines if dependence is detected and if it causes a stall or not

Name Dependence and Output Dependence

- Two instructions use the same name but no flow of information
- Not a true data dependence, but is a problem when reordering instructions

Add R1, R2, R3	**1**	**2**
Sub R3, R5, R6	**2**	**1**

- **Antidependence**: Instruction j writes a register or memory location that instruction I reads

  * Initial ordering (i before j) must be preserved

- **Output dependence**: Instruction i and instruction j write the same register or memory location

  * Ordering must be preserved

Add R1, …, …

Mul R1, …, …

Sub R2, R1, …

Control Dependence

- Ordering of instruction with respect to a branch instruction

  - Instruction that is control dependent on a branch cannot be moved **before** the branch so that its execution is no longer controller by the branch

  - An instruction that is not control dependent on a branch cannot be moved **after** the branch so that its execution is controlled by the branch

```
if p1  {
        S1;
};
If p2 {
        S2;
}
```

Control Dependence

- Instruction that is control dependent on a branch cannot be moved **before** the branch so that its execution is no longer controller by the branch

- An instruction that is not control dependent on a branch cannot be moved **after** the branch so that its execution is controlled by the branch

Example 1:

```
   DADDU R1, R2, R3
   BEQZ  R2, L1
   LW    R1, 0(R2)
L1: …
```

Example 2:

```
   DADDU R1, R2, R3
   BEQZ  R4, skip
   DSUBU R1, R5, R6
skip:
   OR    R7, R1, R8
```

Example 3:

```
   DADDU R1, R2, R3
   BEQZ  R12, skip
   DSUBU R4, R5, R6
   DADDU R5, R4, R9
skip:
   OR    R7, R8, R9
```

Compiler Techniques for Exposing ILP

- Find and overlap sequence of unrelated instruction
- Pipeline scheduling

  - Separate dependent instruction from the source instruction by pipeline latency of the source instruction
- Example:

```
for (i=999; i>=0; i=i-1)
  x[i] = x[i] + S;
```

Instruction producing result	Instruction using result	Latency in clock cycles
FP ALU op	Another FP ALU op	3
FP ALU op	Store double	2
Load double	FP ALU op	1
Load double	Store double	0

Latencies of FP operations. The last column is the number of intervening clock cycles needed to avoid a stall. These numbers are similar to the average latencies we would see on an FP unit. The latency of a floating-point load to a store is 0, since the result of the load can be bypassed without stalling the store. We will continue to assume an integer load latency of 1 and an integer ALU operation latency of 0.

Pipeline Stalls

```
Loop: L.D      F0, 0(R1)
      stall
      ADD.D    F4, F0, F2
      stall
      stall
      S.D      F4, 0(R1)
      DADDUI R1, R1, #-8
      stall (assume integer
load latency is 1)
      BNE      R1, R2, Loop
```

Example:

```
for (i=999; i>=0; i=i-1)
  x[i] = x[i] + S;
```

Instruction producing result	Instruction using result	Latency in clock cycles
FP ALU op	Another FP ALU op	3
FP ALU op	Store double	2
Load double	FP ALU op	1
Load double	Store double	0

Pipeline Scheduling

```
Loop: L.D      F0, 0(R1)
      stall
      ADD.D    F4, F0, F2
      stall
      stall
      S.D      F4, 0(R1)
      DADDUI R1, R1, #-8
      stall (assume integer
load latency is 1)
      BNE      R1, R2, Loop
```

Scheduled code:

```
Loop: L.D      F0, 0(R1)
      DADDUI R1, R1, #-8
      ADD.D    F4, F0, F2
      stall
      stall
      S.D      F4, 8(R1)
      BNE      R1, R2, Loop
```

Instruction producing result	Instruction using result	Latency in clock cycles
FP ALU op	Another FP ALU op	3
FP ALU op	Store double	2
Load double	FP ALU op	1
Load double	Store double	0

Loop Unrolling

```
Loop: L.D     F0, 0(R1)
      ADD.D   F4, F0, F2
      S.D     F4, 0(R1) ;drop DADDUI & BNE
      L.D     F6, -8(R1)
      ADD.D   F8, F6, F2
      S.D     F8, -8(R1) ;drop DADDUI & BNE
      L.D     F10, -16(R1)
      ADD.D   F12, F10, F2
      S.D     F12, -16(R1) ;drop DADDUI & BNE
      L.D     F14, -24(R1)
      ADD.D   F16, F14, F2
      S.D     F16, -24(R1)
      DADDUI  R1, R1, #-32
      BNE     R1, R2, Loop
```

Scheduled code:

```
Loop: L.D     F0, 0(R1)
      DADDUI  R1, R1, #-8
      ADD.D   F4, F0, F2
      stall
      stall
      S.D     F4, 8(R1)
      BNE     R1, R2, Loop
```

Number of live registers?

Loop Unrolling/Pipeline Scheduling

Loop unrolling

```
Loop: L.D     F0, 0(R1)
      ADD.D   F4, F0, F2
      S.D     F4, 0(R1)
      L.D     F6, -8(R1)
      ADD.D   F8, F6,F2
      S.D     F8, -8(R1)
      L.D     F10, -16(R1)
      ADD.D   F12, F10, F2
      S.D     F12, -16(R1)
      L.D     F14, -24(R1)
      ADD.D   F16, F14,F2
      S.D     F16, -24(R1)
      DADDUI  R1, R1, #-32
      BNE     R1, R2, Loop
```

Scheduled unrolled loop

```
Loop: L.D     F0, 0(R1)
      L.D     F6, -8(R1)
      L.D     F10, -16(R1)
      L.D     F14, -24(R1)
      ADD.D   F4, F0, F2
      ADD.D   F8, F6, F2
      ADD.D   F12, F10, F2
      ADD.D   F16, F14, F2
      S.D     F4, 0(R1)
      S.D     F8, -8(R1)
      DADDUI  R1, R1, #-32
      S.D     F12, 16(R1)
      S.D     F16, 8(R1)
      BNE     R1, R2, Loop
```

Strip Mining

Unknown number of loop iterations?

- Goal: make k copies of the loop body (Number of iterations = n)
- Generate pair of loops:

  - First executes n mod k times

  - Second executes n/k tims

  - Strip mining
- Example: Let n=35, k=4

  - Loop 1 executes 3 times (35%4 = 3)

  - Loop 2 executes 8 times by unrolling 4 copies per iteration

Steps in Loop Unrolling and Scheduling

- Determine that unrolling the loop would be useful
- Identify independency of loop iterations
- Use different registers to avoid unnecessary constraints put in on same computations
- Eliminate the extra test and branch instructions and adjust the loop termination and iteration code
- Determine whether the loads and stores from different iterations are independent
- Schedule the code, preserving any dependences needed to yield the same result as the original code

Loop Unrolling and Pipeline Scheduling

Limitations of loop unrolling:

- Code size limitations: I-cache miss
- Compiler limitations: register pressure

Summary

Suggested Readings

Computer Architecture: A Quantitative Approach (5th Edition)
by John Hennessy and David A Patterson:

- pp. 148 - 162: Sections 3.1 and 3.2